

Set

```
In [1]: s = {}
```

```
In [2]: s
```

```
Out[2]: {}
```

```
In [3]: type(s)
```

```
Out[3]: dict
```

```
In [4]: s1 = {1,2,3,3}  
s1
```

```
Out[4]: {1, 2, 3}
```

```
In [5]: s = set()  
type(s)
```

```
Out[5]: set
```

```
In [6]: s1
```

```
Out[6]: {1, 2, 3}
```

```
In [7]: s2 = {100, 1, 34, 9, 43}  
s2
```

```
Out[7]: {1, 9, 34, 43, 100}
```

```
In [8]: s3 = {2, 3.4, True, 1+2j, 'nit'}  
s3
```

```
Out[8]: {(1+2j), 2, 3.4, True, 'nit'}
```

```
In [9]: print(s1)  
print(s2)  
print(s3)
```

```
{1, 2, 3}  
{1, 34, 100, 9, 43}  
{True, 2, 3.4, 'nit', (1+2j)}
```

```
In [10]: s1
```

```
Out[10]: {1, 2, 3}
```

```
In [11]: s1.add(4)  
s1
```

```
Out[11]: {1, 2, 3, 4}
```

```
In [12]: s1.add(5, 6)
s1
```

```
-----
TypeError                                Traceback (most recent call last)
Cell In[12], line 1
----> 1 s1.add(5, 6)
      2 s1

TypeError: set.add() takes exactly one argument (2 given)
```

```
In [13]: s1
```

```
Out[13]: {1, 2, 3, 4}
```

```
In [14]: s4 = s1.copy()
s4
```

```
Out[14]: {1, 2, 3, 4}
```

```
In [15]: s4[0]
```

```
-----
TypeError                                Traceback (most recent call last)
Cell In[15], line 1
----> 1 s4[0]

TypeError: 'set' object is not subscriptable
```

```
In [16]: s4[:]
```

```
-----
TypeError                                Traceback (most recent call last)
Cell In[16], line 1
----> 1 s4[:]

TypeError: 'set' object is not subscriptable
```

```
In [17]: s1
```

```
Out[17]: {1, 2, 3, 4}
```

```
In [18]: s3
```

```
Out[18]: {(1+2j), 2, 3.4, True, 'nit'}
```

```
In [19]: s3.pop()
```

```
Out[19]: True
```

```
In [20]: s3.pop(3)
```

```
-----
TypeError                                Traceback (most recent call last)
Cell In[20], line 1
----> 1 s3.pop(3)

TypeError: set.pop() takes no arguments (1 given)
```

```
In [21]: s1.clear()  
s1
```

```
Out[21]: set()
```

```
In [22]: s1 = {1,2,3,3}  
s1
```

```
Out[22]: {1, 2, 3}
```

```
In [23]: s3.remove(100)
```

```
-----  
KeyError                                Traceback (most recent call last)  
Cell In[23], line 1  
----> 1 s3.remove(100)  
  
KeyError: 100
```

```
In [24]: s3.discard(100)
```

```
In [25]: s3
```

```
Out[25]: {(1+2j), 2, 3.4, 'nit'}
```

```
In [26]: s3.discard(2)  
s3
```

```
Out[26]: {(1+2j), 3.4, 'nit'}
```

```
In [27]: s3.discard(200)  
s3
```

```
Out[27]: {(1+2j), 3.4, 'nit'}
```

```
In [28]: print(s1)  
print(s2)  
print(s3)  
print(s4)
```

```
{1, 2, 3}  
{1, 34, 100, 9, 43}  
{3.4, 'nit', (1+2j)}  
{1, 2, 3, 4}
```

```
In [29]: s3 = {3.4, 1+2j, 'nit'}  
s3
```

```
Out[29]: {(1+2j), 3.4, 'nit'}
```

Set Operation

```
In [30]: a = {1, 2, 3, 4, 5}  
b = {4, 5, 6, 7, 8}  
c = {8, 9, 10}
```

```
In [31]: a.union(b)
```

```
Out[31]: {1, 2, 3, 4, 5, 6, 7, 8}
```

```
In [32]: a | b | c
```

```
Out[32]: {1, 2, 3, 4, 5, 6, 7, 8, 9, 10}
```

```
In [33]: print(a)
         print(b)
         print(c)
```

```
{1, 2, 3, 4, 5}
```

```
{4, 5, 6, 7, 8}
```

```
{8, 9, 10}
```

```
In [34]: a.intersection(b)
```

```
Out[34]: {4, 5}
```

```
In [35]: b.intersection(c)
```

```
Out[35]: {8}
```

```
In [36]: a & b
```

```
Out[36]: {4, 5}
```

```
In [37]: a & b & c
```

```
Out[37]: set()
```

```
In [38]: a.difference(b)
```

```
Out[38]: {1, 2, 3}
```

```
In [39]: b.difference(c)
```

```
Out[39]: {4, 5, 6, 7}
```

```
In [40]: b - c
```

```
Out[40]: {4, 5, 6, 7}
```

```
In [41]: print(a)
         print(b)
         print(c)
```

```
{1, 2, 3, 4, 5}
```

```
{4, 5, 6, 7, 8}
```

```
{8, 9, 10}
```

```
In [42]: a.symmetric_difference(b)
```

```
Out[42]: {1, 2, 3, 6, 7, 8}
```

```
In [43]: b.symmetric_difference(c)
```

Out[43]: {4, 5, 6, 7, 9, 10}

In [44]: `b ^ c`

Out[44]: {4, 5, 6, 7, 9, 10}

In [45]: `s1`

Out[45]: {1, 2, 3}

In [46]: `s5 = frozenset(s1)`

In [47]: `s5`

Out[47]: frozenset({1, 2, 3})

In [48]: `type(s5)`

Out[48]: frozenset

In [49]: `s5.pop(3)`

```
-----  
AttributeError                                Traceback (most recent call last)  
Cell In[49], line 1  
----> 1 s5.pop(3)  
  
AttributeError: 'frozenset' object has no attribute 'pop'
```

In [50]: `print(a)`
`print(b)`
`print(c)`

{1, 2, 3, 4, 5}
{4, 5, 6, 7, 8}
{8, 9, 10}

In [51]: `b.symmetric_difference_update(c)`

In [52]: `b`

Out[52]: {4, 5, 6, 7, 9, 10}

In [53]: `print(a)`
`print(b)`
`print(c)`

{1, 2, 3, 4, 5}
{4, 5, 6, 7, 9, 10}
{8, 9, 10}

In [54]: `a2 = {1,2,3,4,5,6,7,8,9}`
`b2 = {3,4,5,6,7,8}`
`c2 = {10,20,30,40}`

In [55]: `a2.issuperset(b2)`

Out[55]: True

In [56]: `a2.isdisjoint(c2)`

Out[56]: True

In [57]: `a2.isdisjoint(b2)`

Out[57]: False

In [58]: `b2.issubset(a2)`

Out[58]: True

In [59]: `c2.issubset(b2)`

Out[59]: False

In [60]: `a3 = {1,2,3,4,5,6,7,8,9}`
`b3 = {30,40,50,60,70,80}`
`c3 = {10,20,30,40}`

In [61]: `a3.issuperset(b3)`

Out[61]: False

In [62]: `b3.issubset(a3)`

Out[62]: False

In [63]: `b3.isdisjoint(a3)`

Out[63]: True

In [64]: `a3.isdisjoint(b3)`

Out[64]: True

In [65]: `b3.isdisjoint(c3)`

Out[65]: False

Dictionary

In [66]: `d = {1: 'one', 2: 'two'}`

In [67]: `d`

Out[67]: {1: 'one', 2: 'two'}

In [68]: `type(d)`

Out[68]: dict

```
In [69]: d.keys()
```

```
Out[69]: dict_keys([1, 2])
```

```
In [70]: d.values()
```

```
Out[70]: dict_values(['one', 'two'])
```

```
In [71]: d[3] = 'three'
d
```

```
Out[71]: {1: 'one', 2: 'two', 3: 'three'}
```

```
In [72]: d['four'] = 4
d
```

```
Out[72]: {1: 'one', 2: 'two', 3: 'three', 'four': 4}
```

```
In [73]: d.pop(4)
```

```
-----
KeyError                                Traceback (most recent call last)
Cell In[73], line 1
----> 1 d.pop(4)

KeyError: 4
```

```
In [74]: d.pop('four')
```

```
Out[74]: 4
```

```
In [75]: 1 in d
```

```
Out[75]: True
```

```
In [76]: 2 in d
```

```
Out[76]: True
```

```
In [77]: 200 in d
```

```
Out[77]: False
```

Range

```
In [78]: range(5)
```

```
Out[78]: range(0, 5)
```

```
In [79]: list(range(0,5))
```

```
Out[79]: [0, 1, 2, 3, 4]
```

```
In [80]: list(range(10,20))
```

Out[80]: [10, 11, 12, 13, 14, 15, 16, 17, 18, 19]

In [81]: `list(range(10, 50, 5))`

Out[81]: [10, 15, 20, 25, 30, 35, 40, 45]

In [82]: `list(range(10, 50, 5, 2))`

```
-----  
TypeError                                Traceback (most recent call last)  
Cell In[82], line 1  
----> 1 list(range(10, 50, 5, 2))  
  
TypeError: range expected at most 3 arguments, got 4
```

In []: